University Course Management System

Phase 1

Name	ID	Section
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Radwa Hussein Mohamed Saad	20246050	G9
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Dr. Dina Ezzat

TA: Eng. Mohamed Ramadan

1)Elicitation Report:

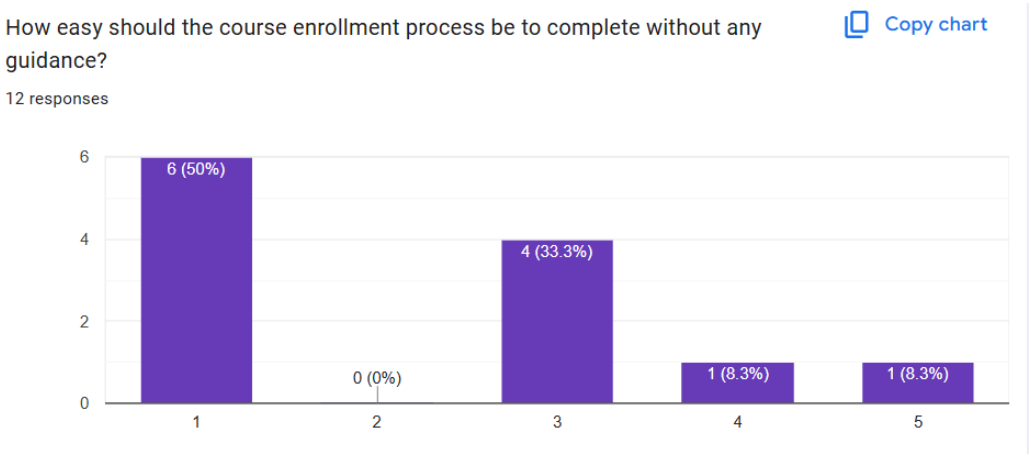
Name: Questionnaire

Justification: A questionnaire was chosen for registered students because they represent a large and distributed user group. A questionnaire allows us to gather responses from a significant number of students efficiently, producing statistical data that reflects the needs and expectations of the majority of the system's users.

Stakeholders Details: 12 registered students from MSA University, spanning various majors. Respondents are primarily second-year students who actively use the university system for course registration and academic activities.

Timing/Location: Questionnaire distributed online via Google Forms on April 17,2026

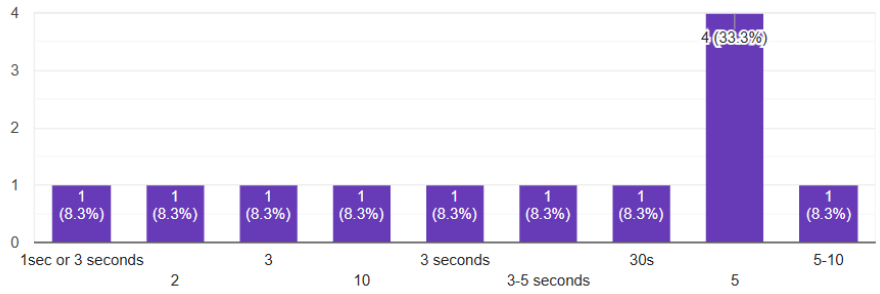
Full Evidence: https://forms.gle/hVwEAZV7eAP9eA6Y6



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After clicking the "Enroll" button for a course, how many seconds should it take for the system to confirm your enrollment and add it to your dashboard? **Just write a number which refers to seconds**

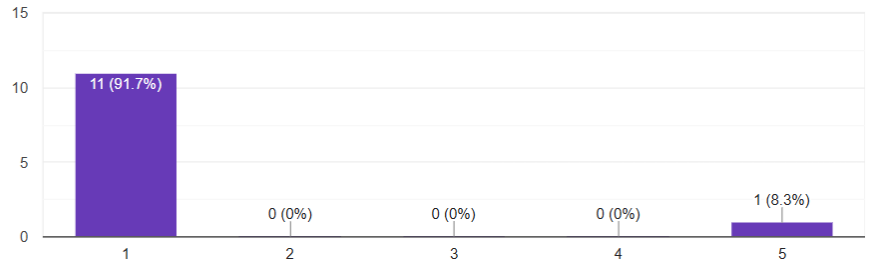
12 responses



Copy chart

How important is it for you to receive an email confirmation after successfully submitting an assignment?

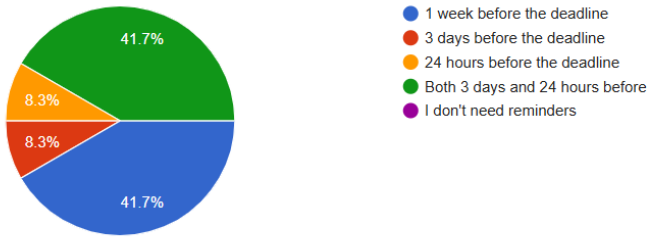
12 responses



Copy chart

When would you most prefer to receive deadline reminder notifications?

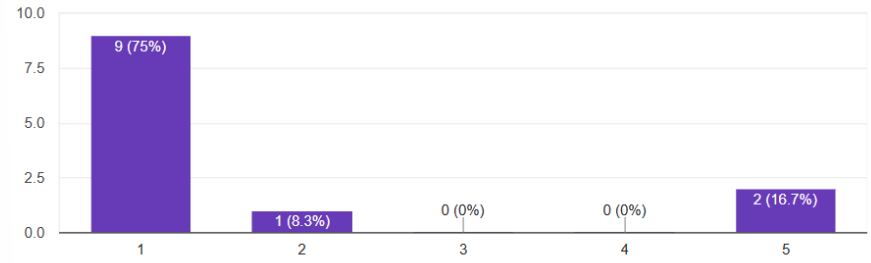
12 responses



How important is it that the assignment creation form saves your progress automatically so you don't lose work if the page crashes?

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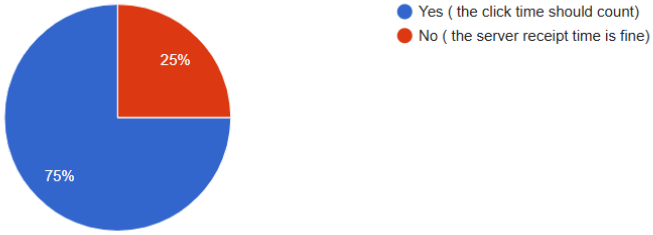
12 responses



If you submit an assignment just before the deadline and the system is slow, would you expect the submission timestamp to reflect when you clicked "Submit" not when the server received it?

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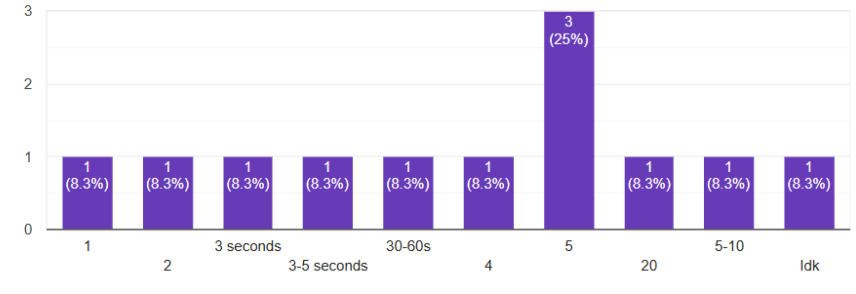
12 responses



When you open your grades page, how many seconds should it take to load all your grades across all enrolled courses before it feels slow? **Just write a number which refers to seconds**

Copy chart

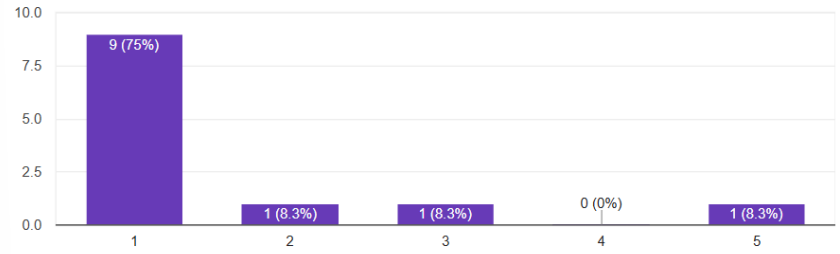
12 responses



How important is it to receive a notification the moment a faculty member posts your grade, so you don't have to keep checking manually?

Copy chart

12 responses



Which of the following grades do you want to see in the system? **(Select all that apply)**

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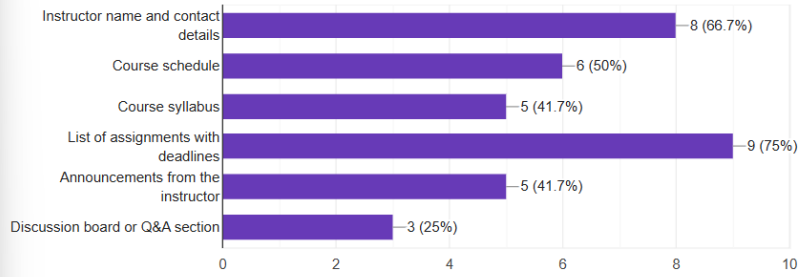
12 responses



When you open a course page, which of the following content items do you expect to find there? **(Select all that apply)**

Copy chart

12 responses



Name: Interview

Justification: Interviews were chosen for faculty members because they are a smaller group with specialized and more complex requirements. A face-to-face interview allowed us to ask follow-up questions, clarify their workflows in depth, and capture detailed information that a questionnaire alone could not provide**.**

Stakeholders Details:

Interviewee 1:

Engineer Lamiaa Atef, Teaching Assistant-Faculty of Computers and Artificial Intelligence, Cairo University

Interviewee 2:

Professor Nisreen Abotalib , Professor of Analytical chemistry -Faculty of Pharmacy, Cairo University

Timing/Location:

Interview 1:

Conducted in person at Cairo University on Saturday, April18,2026

Interview 2:

Conducted online on Sunday, April19,2026

Full Evidence:

1.When creating an assignment, do you prefer to type the assignment details directly within the platform with the deadline or would you rather upload a pre-prepared file (such as a PDF or Word document) containing all the details with deadline?

→ Engineer Lamiaa said that she needs to be able to create an assignment in both ways .

→Professor Nisreen said that uploading file with a deadline is enough.

2.Do you need to assign different assignments to different student groups within the same course?

→ Engineer Lamiaa said NO she do not need to assign.

→Professor Nisreen said Yes she needs to assign.

3.Do you need that one assignment may have multiple deadlines?

→ Engineer Lamiaa said Yes she do needs to.

→Professor Nisreen said Yes she needs to.

4.Do you need to export student progress as a report as PDF?

→ Engineer Lamiaa said Yes she do needs to.

→Professor Nisreen said Yes she needs to.

5.What information do you need to see about a student's progress?

→ Engineer Lamiaa said that she needs information about her course only including quizzes grades , assignments and midterm .

→Professor Nisreen said that she needs information about grades of midterms, finals and GPA of each student in the other courses also information about her course only including quizzes grades , assignments and midterm .

6. When do you want to be notified?

→ Engineer Lamiaa said that she wants to be notified after deadline.

→Professor Nisreen said that she wants to be notified after deadline..

7.Do you need access to the system on mobile, or is desktop enough?

→ Engineer Lamiaa said that she needs to access the system in both ways .

→Professor Nisreen said that she needs to access the system in both ways .

8.How quickly do you expect the system to respond when you perform an action?

→ Engineer Lamiaa said that she needs the system to respond maximum within 2 seconds .

→Professor Nisreen said that she needs the system to respond maximum within 5 seconds .

2)Refined Functional Requirements:

Administrator:

- The system shall allow the administrator to manage user permissions.
- The system shall allow the administrator to create course categories .
- The system shall allow the administrator to update course categories.

- The system shall provide the administrator with tools to monitor, detect, and resolve data inconsistencies across all system records.

Faculty Member:

- The system shall allow the faculty member to create, edit and delete course content.
- The system shall allow the faculty member to create an assignment by either typing details directly on the platform or uploading a pre-prepared file, along with a deadline (Question number 1 in interview)
- The system shall allow the faculty member to set multiple deadlines for a single assignment.(Question number 3 in interview)
- The system shall allow the faculty member to export student progress reports as a PDF file (Question number 4 in interview)
- The system shall allow the faculty member to monitor student progress by viewing assignment submission grades, quizzes grades, and midterm exam of the current course and midterm exam ,final exam, GPA of the other courses. (Question number 5 in interview)
- The system shall generate notifications for faculty member when assignment deadline has passed. (Question number 6 in interview)
- The system shall allow the faculty member to assign different assignments to different student groups within the same course. (Question number 2 in interview)

Registered student:

- The system shall allow the registered student to browse and enroll in available courses.
- The system shall allow the registered student to view instructor name and contact details, course schedule, course syllabus, list of assignments with deadlines, and instructor announcements on course page within the enrolled courses. (Question number 10 in questionnaire)
- The system shall allow the registered student to submit assignments by uploading files before the specified deadline.
- The system shall allow the registered student to view their grades including individual assignment grades, quiz grades, midterm grade, final exam grade, final course grade, cumulative GPA, and GPA per semester also a faculty feedback for all graded assignments within their enrolled courses. (Question number 9 in questionnaire)
- The system shall send an email confirmation to the registered student upon successful assignment submission. (Question number 3 in questionnaire)
- The system shall send deadline reminder notifications to registered students at least 1 week and 24 hours before the assignment deadline. (Question number 5 in questionnaire)
- The system shall automatically save the student's assignment submission progress to prevent data loss in case of a page crash. (Question number 7 in questionnaire)
- The system shall record the assignment submission timestamp as the moment the student clicked Submit,

regardless of server processing delay. (Question number 6 in questionnaire)

- The system shall immediately notify the registered student when a faculty member posts their grade. (Question number 8 in questionnaire)

Guest User:

- The system allow the guest user to view general course information including course title, description, and category without requiring registration or login.

3) Refined Non Functional Requirements:

Performance:

- The system must support 5000 concurrent users without performance degradation
- The system must respond to any user action within a maximum of 2 seconds. (Question number 8 in interview)
- The system shall confirm course enrollment within 5 seconds of the student clicking "Enroll." (Question number 2 in questionnaire)
- The system shall load the full grades page within 5 seconds. (Question number 7 in questionnaire)

Security:

- All sensitive data should be encrypted
- Each user should only access the data permitted for them, 100% of unauthorized access attempts must be blocked

Reliability:

- The System uptime must be 99%

Usability:

- Students must be able to enroll in course in less than 2 minutes
- Faculty members and students must be able to upload assignments in less than 2 minutes
- The course enrollment process must be completable without any guidance or training (Question number 1 in questionnaire)

Portability:

- The system must be fully accessible on both desktop and mobile devices. (Question number 7 in interview)

4)Use Case Table:

Table 1: Enroll in Course

Field	Description
ID	UC-01
Name	Enroll in Course
Actor	Registered Student
Preconditions	Student is logged into the UCMS and has navigated to the course catalog.
Main Flow	1. Student searches for or selects a specific course from the list. 2. Student clicks the "Enroll" button. 3. System validates if the student meets enrollment criteria. 4. System adds the student to the course roster. 5. System displays a confirmation message.
Alternative Flow	**A1: Course is full.** System notifies student the course is at capacity. **A2: Prerequisites not met.** System displays a message stating the required courses.
Postconditions	Student is successfully enrolled and can now access course content.

Table 2: Create Assignment

Field	Description
ID	UC-02
Name	Create Assignment
Actor	Faculty Member
Preconditions	Faculty member is logged in and is the instructor for the specific course.
Main Flow	1. Faculty selects the course and clicks "Create Assignment". 2. Faculty types assignment details (title, instructions). 3. Faculty sets deadlines and assigns to student groups (Include). 4. Faculty saves the assignment. 5. System notifies students of the new assignment.
Alternative Flow	**A1: Invalid Deadline.** System alerts the faculty if the deadline is in the past.
Postconditions	Assignment is live and visible to the enrolled students.

Table 3: Submit Assignment

Field	Description
ID	UC-03
Name	Submit Assignment
Actor	Registered Student
Preconditions	Student is enrolled in the course and the deadline hasn't passed.
Main Flow	1. Student selects the assignment. 2. Student uploads the file. 3. System processes submission (Include). 4. System sends a submission confirmation (Include).
Alternative Flow	**A1: Late Submission.** If the deadline passed, the system might block upload or mark it "Late". **A2: File format error.** System rejects file types not allowed.
Postconditions	Assignment status is updated to "Submitted".

Table 4: Manage User Permissions

Field	Description
ID	UC-04
Name	Manage User Permissions
Actor	Administrator
Preconditions	Administrator is logged in with high-level access privileges.
Main Flow	1. Admin navigates to the "User Management" module. 2. Admin searches for a specific user via ID or email. 3. Admin modifies the user's role (e.g., promoting a Guest to Registered Student). 4. Admin saves the changes. 5. System updates the database and refreshes the user's session permissions.
Alternative Flow	**A1: User Not Found:** System displays an error message if the search criteria return no results.
Postconditions	The user’s access level is immediately updated across the application.

Table 5: View Student Progress

Field	Description
ID	UC-05
Name	View Student Progress
Actor	Faculty Member
Preconditions	Course contains active students with existing grades or submissions.
Main Flow	1. Faculty member selects a course from their dashboard. 2. Faculty clicks on "View Student Progress". 3. System aggregates and displays student names, completion rates, and current grades. 4. Faculty can choose to "Export Progress Report (PDF)" for offline records.
Alternative Flow	**A1: No Data Available:** System displays a placeholder message if no student activities are recorded yet.
Postconditions	Faculty member obtains a clear overview of the academic standing of the class.

5)Use Case Diagram:

